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WHERE STORIES GROW CLAWS

5.0 EDITION

State Machine & Invariant Atlas

A field guide to durable work, transitions, retries, and failure

For implementers, testers, and incident investigators / November 2026

State Machine & Invariant Atlas

Read each entry as trigger, guard, transition, persistence and failure consequence. A UI success message is not authority: database state and the reviewed purchase boundary are.

Contents

Narration guidance and the existing commit boundary
B7 readable books, Bookshelf and reading copies
B6 listening to saved fiction
B5 retained visual collection
B4 editorial transitions
B3 creative, speech and ending transitions
Conventions and terminal outcomes
Start, continue and clarify
Character control and local correction
Path creation and selection
Provider assignment, vault and consent
Authentication and database startup
UI states and regression traces
Optional quality transitions
People and episode transitions
Fourth-wall transitions
Influence transitions
Resistance and memory transitions
Illustrated-path and save transitions

01 Narration guidance and the existing commit boundary

Current context adds source-linked continuity and a post-history next-move gate before the existing reviewed narration request. This is guidance, not a second state engine. Follow/Steer and explicit Act/Say retain their ownership; Ask remains clarification without advancement or effects. One request in default quality Off, up to four with one reviewer or six with both, remains the declared ceiling.

Truth, current attention, knowledge, presence and possession remain distinct. Previous refusal/acceptance and relationship milestones inform the next choice; a purposeful repeated tactic is possible, while a rephrased rejected remedy is not inherently progress. A quiet beat may have no durable effect. An achieved practical goal may finish and an accepted relationship may continue. The narrator can retain meaningful consequences through existing evidenced fact/relationship effects; no new effect opcode or automatic semantic extractor is introduced.

An editorial boundary supplies a source-grounded handoff, never a transition event. Proposals, intentions and attempts stay distinct from performed events and achieved goals. Existing response validation, exact evidence, transactional stale checks, optional review and player-controlled episode ending still govern commits. Passing context tests does not establish that arbitrary model prose obeys these rules.

02 B7 readable books, Bookshelf and reading copies

Gate separates **Make a readable book** from **Keep playing elsewhere**. A book uses the viewed path and inspection moment without activating it. Set title, author credit, language and optional front/back matter. Details are private server drafts and recover after reload; edits invalidate the current preview. Preview shows contents, included cover/placed art, sample prose and exclusions. Empty planned groups, Ask, directions, private references, other paths and audio do not enter the frozen PublicationDocument. Prose itself may contain fictional secrets.

Build frozen edition renders selected formats or all nine locally: DOCX, ODT, RTF, EPUB, PDF, HTML, Markdown, plain text and JSON. Every adapter uses the same immutable source. EPUB places art on a separate page immediately before its prose. A build is a durable job, independent of later story changes. Reopen a retained edition to inspect/cancel a running job. Restart marks abandoned jobs interrupted; navigation/restart never silently rebuilds. A partial failure retains successful formats. Missing/damaged files remain unavailable until an explicit local build; verified identical files can be reused for free. Earlier editions stay frozen.

Bookshelf loads bounded pages of retained books, audio and owner-only story art. Filter the collection, download available formats, play prepared MP3, or return to the source story. Older source, interrupted jobs and unavailable bytes are labelled. Loading the collection starts no renderer, model request or full-story download. Removing a book edition revokes its links and releases only files with no other edition pins. Source fiction remains intact. Private `.inkmorrow5` saves preserve all paths and source images, but exclude prepared books/audio, jobs and reading-link authority. Download prepared editions separately.

Reading links appear only when an administrator configures `PUBLIC_SHARE_BASE_URL` as an HTTP(S) origin and separately provides reachable hosting/allowed-host/proxy configuration. Local export works with sharing unconfigured. Review the exact frozen contents, choose expiry and explicitly create a link. Anyone possessing it can read that copy until expiry/revocation; downloaded copies cannot be retracted. Copy the URL, or select it manually if clipboard access fails. Stored link listings show expiry/revocation, never recoverable bearer tokens. Create a new link if lost.

The new current-fiction adapter stores only a SHA-256 capability hash. Its token travels in a URL fragment and an Authorization header, not a query/path or request log. The unauthenticated adapter serves only the immutable reader-safe document; all owner edition/assets/job APIs remain behind owner/origin/CSRF guards. Reader responses and pages use no-store/noindex/no-referrer, render text safely and make no AI call. Retired authoring and legacy share APIs remain disabled in production. This feature supplies no automatic hosting/deployment or promise of public reachability.

Schema 29 is appended without altering earlier migration source checksums. Source snapshots embed bounded art bytes in one synchronous read cut, independent of later media removal. Initial limits: 96 MiB per source, 1,000 sources / 512 MiB aggregate, 20 queued/running jobs, 10,000 retained jobs, 256 MiB per output and 2 GiB available rendered files. GC marks unreferenced files unavailable before unlink; new pins require availability. No story prose, private state or provider spend is rewritten.

03 B6 listening to saved fiction

Desk and Gallery offer **Listen to saved text**. Choose the configured MP3-capable Narrator voice, Read aloud, Auto-read or an audiobook edition, and a saved passage, path, chapter or volume. **Preview listening plan (free)** freezes that selection and shows segment counts, cached reuse, maximum synthesis calls and exposure. Only selected narrative prose goes to the Narrator: no Ask, private references, owner directions, correction records or sibling paths. Generated prose itself may contain fictional secrets; this is not semantic spoiler filtering.

A new purchase needs consent scoped to that immutable plan. Synthesis is sequential, with no transport retry, recursive splitting or purchased PCM fallback. Read aloud caps at 16 segments; Auto-read/audiobooks at 512. A unit caps at 16 MiB and an edition at 512 MiB. Choose a smaller passage/chapter/volume if preflight rejects the range. There is a 20-job installation queue and a conservative 4 GiB retained-audio quota.

The shared player keeps a labelled source, preparation progress, cancellation and Return action across rooms. Pause/Stop playback differs from **Cancel remaining synthesis**. Voice choice and playback speed are device preferences; changing speed and replaying a cached edition are free. A story/path change pauses Auto-read and requests cancellation of its remaining synthesis. A failed cancellation is reported rather than treated as confirmation. Tap to resume handles browser autoplay refusal without another purchase. Lock clears private playback UI.

Cancellation prevents undispatched units; dispatched work may still cost money and produce reusable cache bytes. Restart interrupts unfinished work without automatic resumption. Review a new plan to reuse completed units and purchase only permitted missing units. A disappeared or corrupt reviewed cache hit fails instead of silently buying a replacement. Retained job status shows known dollars and unknown-cost attempts separately. A late status reply cannot revive a cancelled player display.

Audio editions retain their source and are labelled older when the selected path has moved on. Download the MP3 separately. Private playable saves carry aggregate spend but exclude synthesized audio, listening plans/jobs, credentials and resumable purchase authority. Keep cold installation backups for configured database/media.

04 B5 retained visual collection

Gallery now lists the story-owned image collection in metadata pages of forty, including unplaced uploads and painting results. Thumbnails load lazily; Preview shows the full image, provenance and bounded retained-use metadata. Choose a cover, private reference or ancestral passage before placing an existing image. Placement and removal are local actions; earlier path snapshots and retained collection bytes remain available. New painting is one explicitly reviewed Illustrator purchase. A changed path, cancellation or restart cannot silently attach a late result: validated bytes remain ready and unplaced, and known/unknown spend remains visible. There is no automatic retry or reference-image conditioning.

The existing 400-image story ceiling remains. Every inventory member is an owned retained root; this batch exposes no permanent asset deletion or garbage collection. Playable save v2 carries all owned images and optional inventory descriptions and provenance, but no painting jobs, action keys, consent or provider credentials. Import creates new image identities and a fresh inventory concurrency value.

Visual Library offers selected-reference bundles (`.inkrefs`): worlds, characters and Scribes with normalized images and private lore/motives. Preview names matching existing entries; import always makes independent copies. A repeated identical import action returns the original copies, changed bytes conflict, and deleted results are reported rather than recreated. Bundles omit credentials, purchase journals and image-conditioning permission. They are unencrypted private files, not reader-safe books or full installation backups.

05 B4 editorial transitions

Virtual episode layout → private proposal → free range/target preview → acknowledged command identity → immutable layout revision + local bookkeeping beat + receipt. Failure preserves the proposal. Stale target/base requires comparison; unknown submission requires receipt inspection or an undispatched fence. Editing a private heading may be incomplete; a published layout must be complete and valid.

A closed chapter covers passages after one anchor through another. The final open tail resolves through the selected head and receives future prose. Splitting keeps chronology. Starting another chapter closes the previous tail and creates an empty future tail. Removing a grouping yields explicit fallback ranges, never removed prose. Empty plans are omitted from book output without creating narrative records. Volume assignment follows volume order; overlapping/future/non-narrative anchors fail. These operations change no scene circumstance, knowledge or episode ending.

Fork before adoption → virtual layout; fork after adoption → applicable revision with tail limited to that fork's head. A sibling's future cannot enter the range. Historical outline inspection is read-only and pages pin their original boundary. Save-copy validates/remaps all revision/anchor/producer pointers. A returned book keeps earlier headings; a later build resolves newer layout. Prose editing remains separately gated. Current storage is schema 29; all six PDFs are regenerated at the final checkpoint.

06 B3 creative, speech and ending transitions

A private creative form moves through local edit → acknowledged draft → reviewed exact target/base → acknowledged action key → local immutable revision/receipt. Conflict keeps both proposals. Stale target requires explicit comparison; an unknown action checks its receipt and can be fenced only if undispatched. Earlier proposals copy into a fresh reviewed destination. Lock discards private DOM and late callbacks cannot reopen it. AI reference development adds a distinct candidate state: received → selected-field acceptance → explicit save. Restore is another comparison, with later edits left unchecked.

Scene proposal → accepted narrative consumption is distinct from narrated scene circumstance → established framing with exact evidence. Failure and Ask do not consume the proposal. A saved incomplete first-person/close-third configuration blocks paid preflight until its required speaker/framing narrator/focal is supplied. New stories use developed length; inherited ones retain legacy-short. Pacing does not change the target.

A free narration plan binds target, intent, configuration, models/capacities and all potential stages. Unknown/oversized/expired/changed plans fail before purchase. An actual oversized stage dispatches no further call; an oversized paid candidate retains its charge. Quality remains bounded to 1/4/6 calls, at most one repair. Say occurrence is performed/attempted/interrupted/not-performed. Only performed assembles the exact authorized utterance once. Source segments distinguish speech, aside and ordinary narration; only an eligible exact source can support an effect. Goals transition to achieved/abandoned/failed/superseded; unspecified legacy outcomes never imply achievement. Optional ending records open-ending/vignette/resolved-thread; the latter needs actual achievement. Immediate unchanged ending → append-only undo; later activity requires another path instead. No timer or goal count forces ending.

B3 introduced schema 25; current schema is 29. All six manuals are regenerated together at the final recovery checkpoint.

07 Conventions and terminal outcomes

Local means no model request. It can still need the local server and can fail because of access, validation, stale revision or storage. Paid means transport may incur a provider charge even if no scene is accepted.

A request is not a beat. Several quality calls may produce one accepted beat; zero accepted beats may still carry charges. Reader prose, private state, call metadata and provider output are distinct records.

The common successful sequence is intent, expected revision, durable request, reviewed dispatch, validated proposal, stale check and atomic commit. The common failure sequence preserves the old head and records known/unknown spend. A successful replay returns a prior result; it is not another transition.

Snapshot changes are immutable history. Correcting, retiring, changing a preference or taking a role appends state rather than editing past prose. Rewind creates a new path. It is not destructive truncation.

Tests should assert both what changed and what did not: path head, revision, cast ownership, provider call count, private projection and billing. A passed happy path alone cannot prove at-most-once behaviour under races.

B1 adds device navigation states for Desk, Chronicle, Codex, Gallery and Gate, plus a contribution task. Room changes do not activate a different path, advance fiction revision, clear the typed session draft or purchase generation. Opening the explicit active-path selector's command remains a mutation. Reading cursor and actual graph head remain separate; a local record after the last narrative passage is not a new page. B2 also separates read-only view path, inspection boundary and presentation head from the active mutation target. Draft states are opening, local-only, saving, acknowledged, conflict, prepared submission and discarded/expired. Only acknowledgement permits dispatch. A lost response moves to status inspection, never automatic paid retry; a durable cancellation fence resolves an undispatched unknown. A successful action cannot erase newer edits. Lock invalidates reads and queued writes before clearing private UI.

A pending reply permits read-only page/room navigation. Completion updates current metadata and announces new prose; a reader who moved away from the submitted tail stays in place. Task-close is distinct from request cancellation. An unchanged submitted draft may be cleared on success; a newer revision with identical text is preserved. Lock clears every private room/task buffer immediately, before awaiting server revocation.

08 Start, continue and clarify

Trigger	Guard	Durable result
Start own situation	Valid title/premise/cast/preferences	New game/path, no paid prose
Start curated situation	Known scenario and valid additions	Authored opening, cast and initial facts
Follow	Active episode, current revision, available reviewed plan	One accepted narrated beat
Steer	Valid direction and scope	Beat; ongoing focus only when explicitly selected
Ask	Valid outside-story question	Clarification, no fictional scene advancement
Submit raw opening prose	Production route rejects the field	No game created

A pending purchase blocks conflicting local mutations. Invalid input fails before dispatch. If the selected role is unavailable, the game does not silently choose a different model. A draft response must pass structural validation even when quality is Off.

Ask is still a paid response, but cannot invent an in-world event, grant a challenge, increment the fourth-wall scene cooldown or resolve an episode goal. It is not a hidden-solution endpoint.

Neither starting a story nor reading its existing opening starts a provider call. There is no speculative successor waiting behind Continue. Busy feedback names actual work rather than disguising a second background purchase.

09 Character control and local correction

Inhabit validates a current cast identity and appends a handoff snapshot. Act and Say require that explicit control. Release appends a return to reader-director. None purchases text by itself or replaces the story graph.

Generated effects cannot choose an inhabited person's speech, commitments, decisions or inner relationship expectations. A proposed interaction can leave a choice for the player. Structural guards do not prove every prose sentence respects that ownership; model evaluation remains necessary.

Correction validates an existing public fact or a new fact and a reason, then records an authoritative local change. Earlier prose remains intact. Retirement removes a fact from future retrieval on this path while preserving historical evidence. Neither operation is a memory quota maintenance chore.

A correction and an alternate choice are different transitions. Fix a record that is wrong; fork when exploring another possible history. Local changes do not manufacture narrated payoff simply by marking a goal complete.

Reader evidence omits private reasons. A source link must point to a real ancestor, not a future beat or another path. Import remaps the same constraint to fresh IDs.

10 Path creation and selection

Explore an alternative selects an exact current or earlier moment, checks revision and idle state, then creates a named branch at that snapshot. Rewind uses the same mechanism and preserves the original path.

Selecting an existing branch checks its game ownership and current revision, then changes active selection. Facts, knowledge, resources, commitments, relationships, quality, play style, fourth-wall cooldown, focus, control, episode and placements come from that branch's state.

No branch merge is implied. A promise made after the fork on one side is absent from the other. A reader page outside current ancestry cannot be used as evidence merely because it exists elsewhere in the same story.

At most 40 paths are supported. A limit or stale-revision failure creates no partial path and retains the dialog draft. Refresh before choosing again; never recover a failed local fork by buying narration.

A pending paid request prevents a branch change from becoming a hidden target switch. A late response rechecks exact path and revision before commit. UI route fencing independently prevents an old response from painting another screen.

11 Provider assignment, vault and consent

A role assignment names profile and model. Saving it is local and does not validate quality by buying a completion. Catalogue browsing is a provider read. An unavailable saved model remains unavailable until deliberately repaired.

Session credentials exist in process memory. Vault storage encrypts credentials under a data key wrapped from the owner password. Login can unlock; lock, terminal reset and process disposal remove plaintext access. A remembered browser session after restart does not itself decrypt the vault.

Ordinary paid consent is remembered on the device. Expanded quality uses a separate identity derived from the complete role/provider/model plan and call ceiling. Neither is a substitute for auth, CSRF or the server's revision checks.

A changed plan invalidates purchase authority. Missing memory support cannot be silently omitted from Both. A transport failure does not authorise retry, and a quality rejection does not earn a second repair.

Imported saves contain preferences and aggregate spend, not credentials, consent, call IDs or request keys. Import therefore cannot resume a purchase or authorise a new one. Free local repeated rulings remain independent of provider and quality availability.

12 Authentication and database startup

Static boot is gated. Status resolves setup-needed, locked or unlocked. Setup requires a terminal code and valid password. Login creates an opaque session, its digest is stored, and mutations require the session CSRF token. Lock revokes one session; password change revokes the others.

Private API guards run before body parsing. Before unlock, even a large body cannot become an import, upload or generation job. Production auth is explicit in server.js and cannot be disabled by NODE_ENV=test.

Database startup resolves one file, inspects a private scratch copy of existing database/journals, validates family/version/ledger/integrity, and only then opens the original for normal WAL and transactional migration. Failure cleans only private scratch space, not source data.

Missing/empty source plus orphan journals is rejected. Existing old-family, unknown, corrupt or future databases are not adopted. An accepted 5.0 migration is contiguous and transactional; failure leaves the prior valid version.

After startup, abandoned model requests become interrupted. Their already dispatched uncertain calls remain unknown-cost, not free. No background continuity rebuild, old image queue or retired share service starts.

13 UI states and regression traces

UI state	Allowed action	Required protection
Loading local data	Wait, leave view	Immediate status; late-result fencing
Editing a draft	Edit, cancel, choose input kind	No purchase while typing
Reviewing purchase	Confirm or cancel	Duplicate review guard; retain draft
Submitting	Read busy/progress state	No second paid POST
Failed	Inspect error, free refresh	No automatic paid retry
Locked	Unlock only	Clear private DOM/state and stale callbacks
Ended episode	Read or begin another	No forced continuation

For any network mutation, delay its response, navigate elsewhere, then resolve it. Assert both server accounting and absence of stale UI mutation. Repeat with Lock and a second tab's revision change. Test cancellation before dispatch and failure after a known charge separately.

For uploads/imports, inject validation and transaction failures and inspect staged-file cleanup. For paths and evidence, test sibling/future references. For every optional quality plan, enumerate its maximum calls and the one-repair budget. A passing reviewer must not bypass deterministic state guards.

Use the real production composer without legacyEnabled for boundary tests. Old module fixtures may opt into the retired seam explicitly; their passing tests do not demonstrate that the writing suite is exposed in 5.0.

14 Optional quality transitions

Off: dispatch draft → validate → commit or fail. Enabled: dispatch draft → validate → selected review(s) → commit if approved. An evidenced rejection permits one repair → validate → all selected reviews → commit or fail. An invalid initial draft can consume the same sole repair before review. Invalid reviewer output, transport failure, stale state or changed provider configuration terminates work; none grants another call. One reviewer caps at four total calls, both at six.

Each call moves pending → completed or failed; restart moves pending → interrupted. A late completion may replace interrupted billing uncertainty but cannot reactivate the parent request or append a scene. Dispatched unknown costs are not zero. Known charges remain counted when another call lacks a price. A parent with call rows is excluded from the parent-only spend sum. Exactly one accepted beat and state snapshot commit, regardless of how many draft/review calls were bought.

An unchanged challenge ruling takes its local zero-call path before provider or quality availability checks. Successful idempotency replay also buys nothing. Quality consent is scoped to the server plan; old one-call consent and saved story data cannot authorise an expanded purchase.

15 People and episode transitions

Relationship development: quoted passage/input evidence → existing relationship identity → ownership check → new description with unchanged aspect, people and visibility → immutable change with prior provenance. World-fact replacement is not a develop transition. Affection does not imply trust or cooperation.

Episode opening → public developments → narrated resolution of its recorded goals → payoff → subsequent aftermath. These are descriptive opportunities, not forced story beats. No transition ends the episode: the player explicitly ends it or starts the next one. Quiet turns and local goal corrections do not fabricate a payoff. Rewind restores the earlier snapshot; saves remap payoff evidence and reject missing, cross-path or non-scene payoff references.

Catch me up → public current-path read → bounded existing summaries and records. There is no purchased recap, autonomous absence step or simulated offline progress. Failed path/episode writes retain their local dialog drafts and create no new path.

16 Fourth-wall transitions

Local preference change → snapshot the Never/Rarely/Freely choice, leaving the last-address index intact. Narration → compute permission from play style, participation, eligible non-inhabited cast and scene gap → validate optional aside → append named address and updated index with the complete scene. Rarely permits at most one structured address in six narrated scenes. Failed, stale, Ask and free repeat operations cannot advance the cooldown. No timer runs while away.

A forbidden address fails the response without partial prose or state; actual provider spend remains recorded. Rewind restores the earlier index; save import validates it against scene count and preserves it in the new copy. An aside is neither world-state evidence nor a new source of challenge authority.

17 Influence transitions

Invitation selection -> editable direction only, no request. Moment Steer -> one response, unchanged ongoing focus. Ongoing Steer -> validated response and focus change in one commit. Failure/cancel -> retained draft and scope, unchanged focus. Clear focus -> explicit local preference snapshot. Rewind restores the prior focus and style. Acting, speaking and asking cannot set an ongoing editorial focus.

Challenge review -> local comparison at expected revision -> either ordinary paid consent or reuse of the recorded ruling. A changed revision fails before dispatch. Source navigation -> current-ancestry check -> reader-safe beat, never a private state snapshot. Prior evidence references cannot point to a future or sibling path.

18 Resistance and memory transitions

Explicit approach: validate intent and challenge -> resolve requirements against current-ancestry facts -> determine outcome and stable basis. An unchanged prior decision -> local clarification, zero provider attempts, unchanged fictional scene count. New grounds -> one reviewed narration -> check returned outcome/evidence -> atomic prose, state, adjudication and usage. Wrong structured outcomes fail without a partial commit; known charges remain. Grants remain recorded after later changes to the grounds; rewind restores the previous decision set rather than rerolling it.

Facts: append evidence -> keep immutable change -> retain a bounded working set. Historical retrieval chooses the latest fact version on the active ancestry. Correction supersedes; retirement hides from retrieval; neither deletes history. Forks cannot retrieve a sibling's future. Saves preserve challenge definitions and adjudication evidence, validate ancestry and remap moment identities on import.

19 Illustrated-path and save transitions

Local upload: validate target -> normalize raster -> stage private file -> append placement snapshot and asset metadata atomically. Failure before commit removes only the staged file. Describe/remove appends another snapshot; no prose is edited. Fork/rewind restores the exact placement set at that moment. Historical assets stay.

Paid painting: journal pending -> check reviewed Illustrator -> mark dispatched -> one provider attempt -> validate raster -> compare revision/path -> atomically save asset, placement and terminal usage. Failure/interruption retains known or unknown spend and never retries. Completed keys replay for free even after path changes.

Save: idle story -> complete graph/media snapshot -> bounded gzip file. Import: bounded decompression -> strict graph/media validation -> read-only preview -> explicit copy -> stage new media -> single deferred-FK transaction -> new story. Rollback removes its staged files, never original data. Imported aggregate spend is terminal; request keys, credentials and consent do not travel. Books are a separate read-only active-path projection, with EPUB image-only spine pages before text.

Catalogue creation/editing: validate typed fields -> compare expected revision -> require idle entry -> save metadata and increment revision. No provider call. Upload follows raster normalization and stale recheck; replacement cleans only the previous catalogue image after commit. Catalogue deletion scrubs content and removes its owned image but preserves image accounting. Frozen story copies are not targets.

AI reference development: collect bounded seed fields -> review Storyteller, model, private-field exposure and two-call ceiling -> journal pending by idempotency key -> dispatch once -> validate the complete typed result -> if invalid, dispatch one correction -> return an editable unsaved form. Cancel saves nothing. Failure or restart records known/unknown spend and requires a new explicit take; successful key replay returns the same draft without purchase. Saving and painting remain separate.

Catalogue painting uses its own pending/succeeded/failed/interrupted journal, one pending request per entry and one dispatch per key. A changed provider or revision refuses attachment. Restart marks pending work interrupted; a late completion can settle its known cost but cannot make the image current. There is no silent retry.

Setup selection freezes world, Scribe and character references, copies normalized images to fresh story-owned identities and commits the graph atomically. Story cover and reference updates append path-local snapshots through the story request journal. Saves validate/remap every visual asset reference. Covers enter books; other reference art remains private. Images do not create facts, knowledge or cast control.